Definition 1.19. Given a ring extension B/A, the ring $\tilde{A} = \{b \in B : b \text{ is integral over } A\}$ is the *integral closure of A in B*. When $\tilde{A} = A$ we say that A is *integrally closed in B*. For a domain A, its *integral closure* (or *normalization*) is its integral closure in its fraction field, and A is *integrally closed* (or *normal*) if it is integrally closed in its fraction field.

Proposition 1.20. *If $C/B/A$ is a tower of ring extensions in which B is integral over A and C is integral over B then C is integral over A.*

Proof. See [1, Thm. 10.27] or [2, Cor. 5.4]. ☐

Corollary 1.21. *If B/A is a ring extension, then the integral closure of A in B is integrally closed in B.*

Proof. Let A' be the integral closure of A in B and let A'' be the integral closure of A' in B. Proposition 1.20 implies that A'' is integral over A. Every element of $B \supset A''$ that is integral over A lies in A' (by definition), so $A' \subseteq A'' \subseteq A'$. Thus A' is equal to its integral closure (hence integrally closed) in B. ☐

Proposition 1.22. *The ring $\mathbb{Z}$ is integrally closed.*

Proof. We apply the rational root test: suppose $r/s \in \mathbb{Q}$ is integral over $\mathbb{Z}$, where r and s are coprime integers. Then

$$\left(\frac{r}{s}\right)^n + a_{n-1}\left(\frac{r}{s}\right)^{n-1} + \cdots a_1\left(\frac{r}{s}\right) + a_0 = 0$$

for some $a_0, \ldots, a_{n-1} \in \mathbb{Z}$. Clearing denominators yields

$$r^n + a_{n-1}sr^{n-1} + \cdots a_1 s^{n-1} r + a_0 s^n = 0,$$

thus $r^n = -s(a_{n-1}r^{n-1} + \cdots a_1 s^{n-2} r + a_0 s^{n-1})$ is a multiple of s. But r and s are coprime, so $s = \pm 1$ and therefore $r/s \in \mathbb{Z}$. ☐

Corollary 1.23. *Every unique factorization domain is integrally closed. In particular, every PID is integrally closed.*

Proof. The proof of Proposition 1.22 works for any UFD. ☐

Example 1.24. The ring $\mathbb{Z}[\sqrt{5}]$ is not a UFD (nor a PID) because it is not integrally closed: consider $\phi = (1+\sqrt{5})/2 \in \operatorname{Frac}\mathbb{Z}[\sqrt{5}]$, which is integral over $\mathbb{Z}$ (and hence over $\mathbb{Z}[\sqrt{5}]$), since $\phi^2 - \phi - 1 = 0$. But $\phi \notin \mathbb{Z}[\sqrt{5}]$, so $\mathbb{Z}[\sqrt{5}]$ is not integrally closed.

It follows that every discrete valuation ring is integrally closed. In fact, more is true.

Proposition 1.25. *Every valuation ring is integrally closed.*

Proof. Let A be a valuation ring with fraction field k and let $\alpha \in k$ be integral over A. Then

$$\alpha^n + a_{n-1}\alpha^{n-1} + a_{n-2}\alpha^{n-2} + \cdots + a_1\alpha + a_0 = 0$$

for some $a_0, a_1, \ldots, a_{n-1} \in A$. Suppose $\alpha \notin A$. Then $\alpha^{-1} \in A$, since A is a valuation ring. Multiplying the equation above by $\alpha^{-(n-1)} \in A$ and moving all but the first term on the LHS to the RHS yields

$$\alpha = -a_{n-1} - a_{n-1}\alpha^{-1} - \cdots - a_1\alpha^{2-n} - a_0\alpha^{1-n} \in A,$$

contradicting our assumption that $\alpha \notin A$. It follows that A is integrally closed. ☐